

Automorphism Forms on GL_2

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1 Modular Forms

The map $z \mapsto \begin{bmatrix} 1 & -i \\ 1 & i \end{bmatrix} z$ maps $\mathcal{H}$ isomorphically to D .

Proof: Call this map f . f is well-defined since $|z - i| < |z + i|$ for $z \in \mathcal{H}$. Suppose $f(z) = f(w)$ for $z, w \in \mathcal{H}$. Observe $\begin{bmatrix} 1 & -i \\ 1 & i \end{bmatrix} z \neq 1$, so $\begin{bmatrix} 1 & 1 \\ i & -i \end{bmatrix} \begin{bmatrix} 1 & -i \\ 1 & i \end{bmatrix} z = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} z = z = w$. Suppose $u = x + yi \in D$. Consider $\begin{bmatrix} 1 & 1 \\ i & -i \end{bmatrix} u$.

$$\frac{(x+1) + yi}{i((x-1) + yi)} = \frac{(y + (-x-1)i)((x-1) - yi)}{(x-1)^2 + y^2}.$$

In particular the imaginary part is $\frac{1-x^2-y^2}{(x-1)^2+y^2} > 0$, so $\begin{bmatrix} 1 & 1 \\ i & -i \end{bmatrix} u \in \mathcal{H}$. Then

$\begin{bmatrix} 1 & -i \\ 1 & i \end{bmatrix} \begin{bmatrix} 1 & 1 \\ i & -i \end{bmatrix} u = u$, so f is surjective.